

Registration no:

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Total Number of Pages: 02

B.TECH
PCBM4302

5th Semester Regular / Back Examination 2015-16

SIGNALS AND SYSTEMS

Branch: AEIE, BIOMED, CSE, EC, EIE, ETC, IEE, IT

Time: 3 Hours

Max Marks: 70

Q.Code: T592

**Answer Question No.1 which is compulsory and any five from the rest.
The figures in the right hand margin indicate marks.**

- Q1** Answer the following questions: **(2 x 10)**
- a) What is sampling theorem? Find the Niquist rate of sampling of the signal given below
$$2\sin^2(1500\pi t) - \cos(1750\pi t)$$
 - b) Name three standard discrete signals used for analysis of discrete system. Which is most popular among the three? Justify.
 - c) Find out the step response of a system if the impulse response of the system is $h(n) = u(n - 5)$. Use Z-transform.
 - d) State the properties of discrete time convolution.
 - e) How many real multiplication and addition is required for computation of 32-point DFT?
 - f) What is linearity property of DFT?
 - g) Find DFT of $X^*(n)$?
 - h) Explain the time scaling property of z-transform.
 - i) Find the Fourier transform of $\sin \Omega t$.
 - j) State the properties of autocorrelation and cross correlation of discrete time sequence.
- Q2** a) What is discrete time signal? Differentiate between **(5)**
(i) periodic and aperiodic Signals
(ii) energy and Power signals,
- b) Find the impulse response of the system $h(n)$ described by second order differential equation **(5)**
$$Y(n) = 2y(n - 1) - y(n - 2) + x(n) + 3x(n - 1)$$
- Q3** a) Find the Z-transform of the following signal **(5)**
(I) $x(n) = a^n u(n + 1)$
(II) $x(n) = n^2 u(n - 3)$
- b) Find inverse Z-transform of the casual signal $x(n)$ whose Z-transform is given as **(5)**

$$X(z) = \frac{1}{1 - 1.5z^{-1} - 0.5z^{-2}}$$

- Q4 (a)** Find the Fourier transform of $\text{sgn}(t)$ (5)
- (b)** Find continuous time Fourier transform of a periodic impulse train separated by period T and having amplitude A

- Q5 a)** Prove that DFT of real and even sequence is also real valued and even
- b)** Determine if the system is described by the input output equation $Y(n) = x(n) + 2x(n-1)$ is stable /unstable, Linear/non linear. (5)

- Q5 a)** Find 4-point DFT of the signal, $X(n) = \{1, 1, 0, 1\}$ and sketch magnitude response (5)
- b)** The DFT of $x(n)$ is described as $X(k) = \{1, -1 + 2j, -1, 1 + 2j\}$. Find the DFT of $x^2(n)$ (5)

- Q6 (a)** Determine the range values of parameter a for which the LTI system with impulse response (5)

$$x(n) = \begin{cases} a^n & n \geq 0, \text{ } n \text{ is even} \\ 0 & \text{otherwise} \end{cases}$$

Is stable

- (b)** Prove that convolution of two signals in discrete time domain is equal to multiplication in discrete frequency domain

- Q7 a)** Determine pole-zero plot for the discrete signal (5)
- $$x(n) = \begin{cases} a^n & 0 \leq n \leq M-1 \\ 0 & \text{otherwise} \end{cases}$$

Where, $a > 0$.

- b)** The impulse response of LTI system is expressed as (5)

$$h(n) = 0.2^n u(n)$$

Find the value of A such that $h(n) - A h(n-1) = \delta(n)$

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- Q8** Write short Notes on (Any two) (5 x 2)

- a)** Recursive and non-recursive realization of FIR system
- b)** Aliasing in time and frequency domain
- c)** circular correlation
- d)** Gibb's phenomenon

Registration no:

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Total Number of Pages: 2

B.Tech
PCBM4302

5th Semester Regular / Back Examination 2016-17

SIGNALS AND SYSTEMS

BRANCH(S): CSE, IT, ITE

Time: 3 Hours

Max Marks: 70

Q.CODE: Y381

Answer Question No.1 which is compulsory and any five from the rest.
The figures in the right hand margin indicate marks.

Q1 Answer the following questions: (2 x 10)

- State whether the following discrete time system is static or dynamic where $y(n) = \log_{10}|x(n)|$
- State whether the following system is BIBO stable or not where $y(n) = x(n) - x(-n-1) + x(n-1)$
- Prove that $a_0 = 0$ and $a_n = 0$ (where a_0 and a_n represents trigonometric Fourier Series Coefficients) if $x(t)$ exhibits odd symmetry.
- Find the Fourier transform of A (constant).
- Find the Z transform of $x^*(n)$.
- Determine the energy and power of a d.t. unit step signal.
- Find the DFT of $\delta(n)$.
- Find the DFT of a purely real and even signal $x(n)$.
- Distinguish between FIR and IIR systems.
- Prove that autocorrelation at zero lag represents energy of a discrete time signal $x(n)$.

Q2 Define autocorrelation and cross correlation. Show that autocorrelation is an even function of lag 'l' and it is maximum at zero lag. (2+8)

Q3 a) Prove that the Parseval's relation in the Z domain (5)

$$\sum_{-\infty}^{\infty} x_1(n)x_2^*(n) = \frac{1}{2\pi j} \int X_1(Z)X_2^*\left(\frac{1}{Z^*}\right)Z^{-1}dZ$$

- b) Find the linear convolution between $x_1(n) * x_2(n)$ where (5)**
 $x_1(n) = \{1, 2, 3, 4\}$ and $x_2(n) = \{-1, 0, 1, 2\}$ using graphical method.

Q4 a) Perform a 4 point circular convolution using DFT/IDFT method where $x_1(n) = \{-1, 2, -3, 4\}$ and $x_2(n) = \{4, 7, 2\}$ (5)

b) Determine the inverse Z transform of $X(Z) = \ln \frac{z}{z-1}$ ROC $|z| > \frac{1}{e}$ (5)

Q5 a) Determine all possible values of $x(n)$ for $X(Z) = \frac{z^3 + z^2 + 1.5z + 0.5}{z^3 + 1.5z^2 + 0.5z}$ (5)

b) An LTI system is described by the difference equation $y(n) = x(n) + 0.81x(n-1) - 0.81x(n-2) - 0.45x(n-2)$. Determine the system function of the system. Sketch the poles and zeros on the Z plane. Assess the stability. (5)

Q6 a) Solve the following difference equation using unilateral Z transform (5)

$$y(n) - \frac{7}{12}y(n-1) + \frac{1}{12}y(n-2) = x(n) \text{ for } n \geq 0 \text{ with initial conditions } y(-1) =$$

$$2; y(-2) = 4 \text{ and } x(n) = \left(\frac{1}{5}\right)^n u(n)$$

b) Define circular convolution. Derive the circular convolution relation between $x_1(n)$ and $x_2(n)$ (5)

Q7 Find the Fourier transform of periodic impulse train having amplitude 1 and time period T_0 . Find the trigonometric form of Fourier series of the half wave rectified sine wave having peak amplitude A (10)

Q8 Write short answer on any TWO: (5 x 2)

a) Gibb's phenomena

b) Sampling theorem

c) Parseval's relation

d) Stability analysis in the Z domain

Registration No:

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Total Number of Pages: 3

B.TECH
PET31103

3rd Semester Regular Examination 2016-17

SIGNAL & SYSTEMS

BRANCH(S): ECE, ETC

Time: 3 Hours

Max Marks: 100

Q.CODE: Y603

Answer Part-A which is compulsory and any four from Part-B.
The figures in the right hand margin indicate marks.

Part – A (Answer all the questions)

Q1 Answer the following questions: *dash fill up type* (2 x 10)

- a) All periodic signals are: ____ (energy/power/neither energy nor power)
- b) The even part of the signal $x(n) = e^{j\frac{\pi}{3}n}$ is ____.
- c) The Twiddle factor $W_N =$ ____.
- d) Autocorrelation function is ____ function.
- e) Check the linearity of $y(n) = Ax(n) + B$: ____
- f) For finite anti-causal sequence the ROC in Z-transform is ____.
- g) The normalized cross-correlation sequence $\rho_{xy}(l) =$ ____.
- h) The exponential form of Fourier series of a periodic signal $x(t)$ with period T is defined by ____.
- i) The system function described by the difference equation $y(n) = \frac{1}{2}y(n-1) + 2x(n)$ is ____.
- j) If the DFT of $x(n) = X(K)$, then $DFT\{x(n-m)_N\}$ is ____.

Q2 Answer the following questions: *Short answer type* (2 x 10)

- a) Describe all-zero system with an example.
- b) Describe elementary signals: unit sample sequence and ramp signal.
- c) Determine whether $x(n) = u(n)$ is energy or power signal, where $u(n)$ stands for unit step sequence.
- d) Determine whether $x(n) = \sin\left(\frac{6\pi}{7}n + 1\right)$ is periodic or not. If periodic find fundamental period.
- e) Define time-reversal property in Z-transform with its ROC.
- f) State Parseval's relation in Z-transform domain.
- g) Define zero padding and its importance.
- h) State and prove time shifting property of Fourier transform.
- i) Prove $[x_1(n) * x_2(n)] = [x_2(n) * x_1(n)]$, i.e. convolution for LTI system is commutative.
- j) Plot the discrete time sequence $y(n) = -2u(-n-2)$, where $u(n)$ is unit step sequence.

Part – B (Answer any four questions)

Q3 a) Draw the direct form-I and form-II realization of the LTI system given as $2y(n) + y(n-1) - 2y(n-3) = x(n-1) + x(n-3)$. **(10)**

b) State the relationship between convolution and correlation and prove it. **(5)**

Q4 a) Determine the response of the LTI system whose input $x(n]$ and impulse response $h(n]$ are given by, $x(n] = \left\{ \underset{\uparrow}{1}, 2, 3, 1 \right\}$ and $h(n] = \left\{ 1, \underset{\uparrow}{2}, 1, -1 \right\}$. Use graphical method to solve. **(10)**

b) Prove that for convolution of LTI system the distributive property holds. **(5)**

Q5 a) Derive convolution sum of LTI system and state its properties. **(10)**

b) Show that the necessary and sufficient condition for a relaxed LTI system to be BIBO stable is $\sum_{n=-\infty}^{\infty} |h(n)] \leq M_h < \infty$, for some constant M_h . **(5)**

Q6 a) Determine the response of LTI system governed by the difference equation, $y(n] - 0.5y(n-1] = x(n]$, for input $x(n] = (5)^n u(n]$, and initial condition $y(-1] = 2$. Use one-sided z-transform property. **(10)**

b) Find the Z-transform with its ROC for $n^2 u(n] - 3^n u(-n-1]$. **(5)**

Q7 a) Find the inverse Z-transform of the following **(10)**

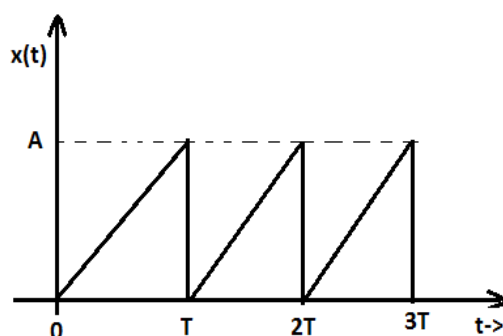
$$X(z) = \frac{1}{(1+z^{-1})(1-z^{-1})^2}$$

using partial-fraction expansion method.

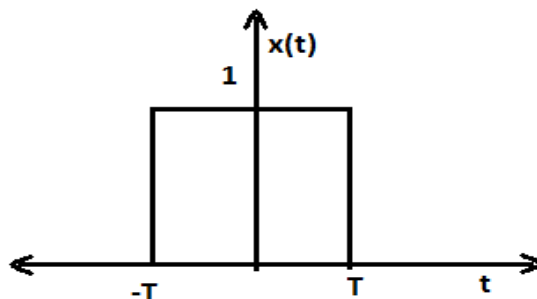
b) Compute 4-point DFT of the causal three sample sequence given by **(5)**

$$x(n] = \begin{cases} \frac{1}{3} & : 0 \leq n \leq 2 \\ 0 & : \text{otherwise} \end{cases}$$

Q8 a) Determine the Trigonometric form of Fourier series of the ramp signal shown below **(10)**



- b) Determine the Fourier transform of rectangular pulse shown below (5)



- Q9 a) Determine the total response $y(n), n \geq 0$ of the system described by the second order difference equation $y(n) - 2y(n-1) - 3y(n-2) = x(n) + 4x(n-1)$, when the input signal is $x(n) = 2^n u(n)$ and with initial condition $y(-2) = 0, y(-1) = 5$. (10)
- b) Consider the following two sequences $x_1(n) = \{2, 3, 1, 4\}$ and $x_2(n) = \{5, 2, 1\}$. Find the circular convolution of $x_1(n)$ and $x_2(n)$. (5)

Registration No :

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Total Number of Pages : 02

B.Tech
PCBM4302

5th Semester Back Examination 2018-19

SIGNALS AND SYSTEMS

BRANCH : AEIE, BIOMED, BIOTECH, CSE, ECE, EIE, ETC, IEE, IT, ITE

Time : 3 Hours

Max Marks : 70

Q.CODE : E544

Answer Question No.1 which is compulsory and any FIVE from the rest.

The figures in the right hand margin indicate marks.

Q1 Answer the following questions : (2 x 10)

- a) Determine whether or not the signal $x[n]$ is periodic. If $x[n]$ is periodic, determine its fundamental time period.

$$x[n] = \cos^2 \frac{\pi}{8} n$$

- b) Determine whether the signal $x[n] = 2e^{j3n}$ is energy signal, power signal nor neither.

- c) Is the system $y(t) = x(2t)$ time invariant?

- d) State and explain frequency shifting property of Fourier series.

- e) Determine the Fourier transform of unit impulse.

- f) Determine appropriate Fourier representations of the signal

$$x(t) = |\sin(2\pi t)|$$

- g) Write conditions for stability and causality of the system with rational transfer function (z) .

- h) State initial value theorem & final value theorem in Z transform.

- i) Determine the z-transform and ROC of the signal

$$x(n) = \delta[n - k], \quad k > 0$$

- j) Write down the circular time shift and circular frequency shift property of DFT.

Q2 a) The impulse response of a discrete-time LTI system is given as (5)

$$h(n) = \left(-\frac{1}{2}\right)^n u(n) + (1.01)^n u(n-1)$$

Determine whether the system is causal and/ or stable.

b) Find the convolution sum (5)

$$y[n] = (u[n+10] - 2u[n] + u[n-4]) * u[n-2]$$

Q3 a) Compute the autocorrelation of the signal $x(n) = \alpha^n u(n)$, $0 < \alpha < 1$ (5)

b) Find the FT of the signal $x(t) = \frac{d}{dt} \{ (e^{-3t} u(t)) * (e^{-t} u(t-2)) \}$ (5)

Q4 a) Find out the FS coefficients by method of inspection for the signal (5)

$$x(t) = \sin(3\pi t) + \cos(4\pi t)$$

b) Find the FT of signal $x(t) = \frac{1}{1+t^2}$ (5)

Q5 a) Suppose we are given the following information about a continuous-time periodic signal with period 3 and Fourier coefficients $X(k)$: (5)

(1) $X(k) = X(k+2)$

(2) $X(k) = X(-k)$

(3) $\int_{-0.5}^{0.5} x(t) dt = 1$

(4) $\int_{0.5}^{1.5} x(t) dt = 2$

Determine (t) .

b) Find the inverse Z-transform

(5)

$$X(z) = \frac{2 + z^{-1}}{1 - \frac{1}{2}z^{-1}}$$

With ROC $|z| > \frac{1}{2}$, using power series expansion

Q6 a) A causal system has input $x[n]$ and output $y[n]$. Use the transfer function to determine the impulse response of this system.

(5)

$$x[n] = \delta[n] + \frac{1}{4}\delta[n-1] - \frac{1}{8}\delta[n-2]$$

$$y[n] = \delta[n] - \frac{3}{4}\delta[n-1]$$

b) Use the unilateral z-transform to determine the forced response, the natural response, and the complete response of the systems described by the following difference equations with the given inputs and initial conditions.

(5)

$$y[n] - \frac{1}{3}y[n-1] = 2x[n], \quad y[-1] = 1, \quad x[n] = \left(-\frac{1}{2}\right)^n u[n]$$

Q7 Perform the circular convolution of the following two sequences $x_1(n)$ & $x_2(n)$. Illustrate the sequences of circular convolution with neat sketches. By means of the DFT and IDFT, determine the sequence $x_3(n)$ corresponding to the circular convolution of the sequences $x_1(n)$ & $x_2(n)$.

(10)

$$x_1(n) = \{2, 1, 2, 1\}, \quad x_2(n) = \{1, 2, 3, 4\}.$$



Q8 Write short answer on any TWO :

(5 x 2)

- a) Duality property of FT
- b) Causality and stability of LTI system in Z domain
- c) Properties of Convolution
- d) Parseval's Theorem

Registration No :

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Total Number of Pages : 02

B.Tech
PET3I103

3rd Semester Regular / Back Examination 2018-19

SIGNAL & SYSTEMS

BRANCH : ECE, ETC

Time : 3 Hours

Max Marks : 100

Q.CODE : E938

Answer Question No.1 (Part-1) which is compulsory, any eight from Part-II and any two from Part-III.

The figures in the right hand margin indicate marks.

Part- I

Q1 Short Answer Type Questions (Answer All-10) (2 x 10)

- List the ROC properties of Laplace transform.
- Find the Z-transform of a sequence $x[n] = \cos(n\omega T) u[n]$.
- Find the Fourier transform of $x(t) = e^{-at}u(t)$
- Is there be two different signals having same Laplace transform? Give an example. How do you differentiate these two signals?
- Define the convolutional integral.
- What is aliasing?
- Given $x(n) = \{1, -4, 3, 1, 5, 2\}$. Represent $x(n)$ in terms of weighted shifted impulse function.
- State the multiplication property of DTFT.
- Using Z-transform check whether the following system is stable or not?

$$H(z) = \frac{z}{z - 1/2} + \frac{2z}{z + 3}, \quad \frac{1}{2} < |z| < 3$$

- State and prove the time folding property of Z- transform.

Part- II

Q2 Focused-Short Answer Type Questions- (Answer Any Eight out of Twelve) (6 x 8)

- Check the system $y(n) = \log_{10}(x[n])$ is linear, time invariant, causal and static.
- Find out whether the following signal is energy or power signal or neither any of two. Determine power or energy if found.
 $x(t) = u(t) + 5u(t - 1) - 2u(t - 2)$
- Find the spectrum of $x(t) = e^{-2|t|}$. Plot the spectrum of the signal.
- Find the Fourier transform of a rectangular pulse with width T and amplitude A.
- State and prove Parseval's theorem of Fourier Transform.
- Find the inverse Laplace transform of $X(s) = \frac{1}{(s+5)(s-3)}$ for the ROC $-5 < \text{Re}\{s\} < 3$
- Using Laplace transform of $x(t)$. Give the pole-zero plot and find ROC of the signal $x(t)$.
 $x(t) = e^{-b|t|}$ for both $b > 0$ and $b < 0$.
- Find the Z-transform and sketch the ROC of the following sequence
 $x[n] = 2^n u[n] + 3^n u(-n - 1)$

- Using the properties of inverse Z-transform solve

$$X(z) = \log(1 + az^{-1}); |z| > |a| \text{ and } X(z) = \frac{az^{-1}}{(1 - az^{-1})^2}; |z| > |a|$$

- Determine DTFT of $\left(\frac{1}{2}\right)^n u(n)$. Plot its spectrum.

- Using convolution property of DTFT, find the inverse DTFT of

$$X(e^{j\omega}) = \frac{1}{(1 - ae^{j\omega})^2}, |a| < 1$$

- Find the convolution sum of $x[n] = r[n]$ and $h[n] = u[n]$.

Part-III

Long Answer Type Questions (Answer Any Two out of Four)

- Q3

Using convolution integral, determine the response of a CTLTI system $y(t)$ given input $x(t) = e^{-\alpha t}u(t)$ and impulse response $h(t) = e^{-\beta t}u(t)$, $|\alpha| < 1, |\beta| < 1$

(16)
- Q4

Analyze on recursive and non-recursive systems with an example

(16)
- Q5

a) Using graphical method, find the output sequence $y[n]$ of the LTI system whose response $h[n]$ is given and input $x[n]$ is given as follows. $x[n] = \{0,5,2\}$ and $h[n] = \{1,1,1\}$.

b) Consider an analog signal $x(t) = 5 \cos 200\pi t$.

i. Determine the minimum sampling rate to avoid aliasing.

ii. If sampling rate is 400Hz, What is the DT signal after sampling?

(16)
- Q6

a) State and prove Sapling Theorem.

b) State and prove following properties of DTFT.

i. Differentiation in frequency

ii. Convolution in frequency domain

(16)

Registration No :

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Total Number of Pages : 02

B.Tech
PET31103

3rd Semester Back Examination 2019-20

SIGNAL & SYSTEMS

BRANCH : ECE, ETC

Max Marks : 100

Time : 3 Hours

Q.CODE : HB895

Answer Question No.1 (Part-1) which is compulsory, any EIGHT from Part-II and any TWO from Part-III.

The figures in the right hand margin indicate marks.

Part- I

Q1 Only Short Answer Type Questions (Answer All-10) (2 x 10)

- Define energy and power signal.
- Find whether the following system with impulse response is $h(t)$ is stable or not. $h(t) = te^{-t}u(t)$.
- Write the condition for stability of a DT-LTI system with respect to the position of poles.
- Give Parseval's relation for continuous Time Fourier Transform.
- State and prove time folding property of z-transform.
- Will there be two different signals having same Laplace transform? Give an example. How do you differentiate these two signals ?
- State the significance of impulse response.
- What is aliasing?
- State multiplication property of DTFT.
- What is the difference between recursive and non-recursive system?

Part- II

Q2 Only Focused-Short Answer Type Questions- (Answer Any Eight out of Twelve) (6 x 8)

- Find the even and odd component of the signal $x(n) = \{1, 0, -1, 2, 3\}$
- Find out whether the following signals are periodic or not. If periodic find the period.
 - $x(t) = 2 \cos(10t + 1) - \sin(4t - 1)$
 - $x(n) = \cos(0.1\pi n)$
- Using Laplace transform of $x(t)$. Give the pole-zero plot and find ROC of the signal $x(t)$. $x(t) = e^{-b|t|}$ for both $b > 0$ and $b < 0$.
- Obtain the Fourier coefficient and write the quadrature form of a fully rectified sine wave.
- Find out whether the following signal is energy or power signal or neither any of two. Determine power or energy if found.
$$x(t) = u(t) + 5u(t - 1) - 2u(t - 2)$$
- What is the transfer function and the impulse response of RC low pass circuit?
- Determine the unit step response of the LTI system defined by
$$d^2y/dt^2 + 5dy/dt + 6y(t) = dx/dt + x(t).$$
- State and prove any two properties of continuous Time Fourier Transform.
- Find the convolution of sum of $x[n] = r[n]$ and $h[n] = u[n]$
- obtain the parallel realization of the system given by
$$y(n) - 3y(n-1) + 2y(n-2) = x(n).$$
- Determine DTFT of $\left(\frac{1}{2}\right)^n u(n)$. Plot its spectrum.
- Using the properties of inverse Z-transform solve

$$X(z) = \log(1 + az^{-1}); |z| > |a| \text{ and } X(z) = \frac{az^{-1}}{(1 - az^{-1})^2}; |z| > |a|$$

Part-III

Only Long Answer Type Questions (Answer Any Two out of Four)

- Q3** Using graphical method, find the output sequence $y[n]$ of the LTI system whose response $h[n]$ is given and input $x[n]$ is given as follows. $x[n] = \{0, 5, 2\}$ and $h[n] = \{1, 1, 1\}$. (16)
- Q4** For a causal LTI system the input $x(n)$ and output $y(n)$ are related through a difference equation $y(n) - \frac{1}{6}y(n-1) - \frac{1}{6}y(n-2) = x(n)$. Determine the frequency response $H(e^{j\omega})$ and the impulse response $h(n)$ of the system. (16)
- Q5** a) It is given that the state matrices for a discrete time system are $A = \begin{bmatrix} 0 & 1 \\ 2 & 3 \end{bmatrix}$, $B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$, $C = [8 \quad 8]$, $D = [1]$. Find the system transfer function. (8)
- b) Find DTFT of $x(n) = [0, 1, 2, 1, 0]$ (8)
- Q6** a) Consider an analog signal $x(t) = 5 \cos 200\pi t$. (8)
- a. Determine the minimum sampling rate to avoid aliasing.
- b. If sampling rate $F_s = 400$ Hz. What is the DT signal after sampling?
- b) Determine the Fourier transform for double exponential pulse whose function is given by $x(t) = e^{-a|t|}$, $a > 0$. Also draw its amplitude and phase spectra. (8)

Registration No :

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Total Number of Pages : 02

B.Tech
REC3C002

3rd Semester Regular Examination 2019-20

SIGNALS AND SYSTEMS

BRANCH : ECE, ETC

Max Marks : 100

Time : 3 Hours

Q.CODE : HR880

Answer Question No.1 (Part-1) which is compulsory, any EIGHT from Part-II and any TWO from Part-III.

The figures in the right hand margin indicate marks.

Part- I

Q1 Only Short Answer Type Questions (Answer All-10) (2 x 10)

- Define energy and power signal.
- Find whether the following system with impulse response is $h(t)$ is stable or not. $h(t) = te^{-t}u(t)$.
- Write the condition for stability of a DT-LTI system with respect to the position of poles.
- Give Parseval's relation for continuous Time Fourier Transform.
- State and prove time folding property of z-transform.
- Will there be two different signals having same Laplace transform? Give an example. How do you differentiate these two signals ?
- State the significance of impulse response.
- What is aliasing?
- State multiplication property of DTFT.
- What is the difference between recursive and non-recursive system?

Part- II

Q2 Only Focused-Short Answer Type Questions- (Answer Any Eight out of Twelve) (6 x 8)

- Find the even and odd component of the signal $x(n) = \{1, 0, -1, 2, 3\}$
- Find out whether the following signals are periodic or not. If periodic find the period.
 - $x(t) = 2 \cos(10t + 1) - \sin(4t - 1)$
 - $x(n) = \cos(0.1\pi n)$
- Using Laplace transform of $x(t)$. Give the pole-zero plot and find ROC of the signal $x(t)$. $x(t) = e^{-b|t|}$ for both $b > 0$ and $b < 0$.
- Obtain the Fourier coefficient and write the quadrature form of a fully rectified sine wave.
- Find out whether the following signal is energy or power signal or neither any of two. Determine power or energy if found.
$$x(t) = u(t) + 5u(t - 1) - 2u(t - 2)$$
- What is the transfer function and the impulse response of RC low pass circuit?
- Determine the unit step response of the LTI system defined by
$$d^2y/dt^2 + 5dy/dt + 6y(t) = dx/dt + x(t).$$
- State and prove any two properties of continuous Time Fourier Transform.
- Find the convolution of sum of $x[n] = r[n]$ and $h[n] = u[n]$
- obtain the parallel realization of the system given by
$$y(n) - 3y(n-1) + 2y(n-2) = x(n).$$
- Determine DTFT of $\left(\frac{1}{2}\right)^n u(n)$. Plot its spectrum.
- Using the properties of inverse Z-transform solve

$$X(z) = \log(1 + az^{-1}); |z| > |a| \text{ and } X(z) = \frac{az^{-1}}{(1 - az^{-1})^2}; |z| > |a|$$

Part-III

Only Long Answer Type Questions (Answer Any Two out of Four)

- Q3** Using graphical method, find the output sequence $y[n]$ of the LTI system whose response $h[n]$ is given and input $x[n]$ is given as follows. $x[n] = \{0, 5, 2\}$ and $h[n] = \{1, 1, 1\}$. (16)
- Q4** For a causal LTI system the input $x(n)$ and output $y(n)$ are related through a difference equation $y(n) - \frac{1}{6}y(n-1) - \frac{1}{6}y(n-2) = x(n)$. Determine the frequency response $H(e^{j\omega})$ and the impulse response $h(n)$ of the system. (16)
- Q5** a) It is given that the state matrices for a discrete time system are $A = \begin{bmatrix} 0 & 1 \\ 2 & 3 \end{bmatrix}$, $B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$, $C = [8 \quad 8]$, $D = [1]$. Find the system transfer function. (8)
- b) Find DTFT of $x(n) = [0, 1, 2, 1, 0]$ (8)
- Q6** a) Consider an analog signal $x(t) = 5 \cos 200\pi t$. (8)
- a. Determine the minimum sampling rate to avoid aliasing.
- b. If sampling rate $F_s = 400$ Hz. What is the DT signal after sampling?
- b) Determine the Fourier transform for double exponential pulse whose function is given by $x(t) = e^{-a|t|}$, $a > 0$. Also draw its amplitude and phase spectra. (8)

Registration No. :

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Total Number of Pages : 02

BTech
REC3C002

3rd Semester Regular / Back Examination: 2021-22

SIGNALS & SYSTEMS

BRANCH(S): ETC / ECE

Time : 3 Hour

Max Marks : 100

Q.Code : OF802

Answer Question No.1 (Part-1) which is compulsory, any eight from Part-II and any two from Part-III.

The figures in the right hand margin indicate marks.

Part-I

Q1 Answer the following questions : (2×10)

- Plot the signal $x(n) = 3u(n) - 2u(n-2) + 2$
- What do you mean by BIBO stable?
- What is normalized Autocorrelation sequence?
- Explain the associative property of linear convolution?
- State the duality property of Fourier transform?
- Write the trigonometric form of Fourier series of a periodic signal?
- Define ROC for one sided Z-transform of a signal?
- Find inverse Z-transform of the signal $X(z) = 5z^{-1} + 6z^{-4} - 8z^3 + 10$
- $x(n) = \{19, 8\}$, find its DFT ?
- What is DTFT and DFT, mention the relationship?

Part-II

Q2 Only Focused-Short Answer Type Questions- (Answer Any Eight out of Twelve) (6×8)

- What is an energy signal, power signal, explain with examples?
- $y(n) = 3x(n) + 5$, is a LTI system or not, Justify ?
- Find the linear convolution between the sequences $\{2, 1, -3\}$ and $\{10, 8, 2\}$
- Find the cross correlation of the sequence $\{-2, 6, 7, 1, 9\}$ and $\{1, -2, 1\}$
- Write and prove the integration property of Fourier Series of a continuous time periodic signal?
- Find the Fourier transform of a triangular signal, with amplitude 5 v and duration 2T, symmetric about Y-axis.
- State and prove the correlation property of the Z-transform.
- $x(n) = \{2, 3, -4, 1, 5, 6, 2, 1\}$, find one sided Z-transform of $x(n+3)$?
- Find the Auto correlation of the sequence $\{1, 2, 5, 2, -1, 4\}$ using Z-transformation.
- Explain symmetry property of DFT.
- Find the DFT of the sequence $\{2+3j, 1, 2-2j, -3\}$
- Derive the condition of stability and causality in Z-domain.

Part-III

Only Long Answer Type Questions (Answer Any Two out of Four)

- Q3

Classify discrete time signals and systems in detail with examples.

(16)
- Q4

What is unit step response of a system? Find the impulse response of a LTI system described by the difference equation $y(n) = x(n) + 2x(n-3)$.

(16)
- Q5

Explain the various methods to find the inverse Z-transform of a signal, with merits and demerits?

(16)
- Q6

Given $x(n) = \{1, -2, 3, 2\}$, $y(n) = \{2, -1, 1, 2\}$, find its circular convolution using DFT, IDFT method ?

(16)

Registration No: -

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Total Number of Pages: 02

Course: B.Tech
Sub Code: REC3C002

3rd Semester Regular / Back Examination: 2022-23

SUB: SIGNALS AND SYSTEMS

BRANCH(S): ECE, ETC

MAX MARKS: 100

TIME: 3 HOURS

Q CODE: L516

Answer Question No.1 (Part-1) which is compulsory, any eight from Part-II and any two from Part-III.

The figures in the right hand margin indicate marks.

Part- I

Q1 Answer All-10

(02x10)

- Find Z-transform of $u(n-1)$.
- Differentiate between autocorrelation and cross correlation.
- What is the superposition theorem to check linearity of a discrete time system?
- How to check causality and stability of a discrete time system?
- Write the formulation for continuous time Fourier transform.
- What is the relation between DFT and z-transform?
- What is the long division method in inverse Z-Transform?
- Determine the DFT of $2\delta(n)+3$.
- Write any two properties of z-transform.
- Differentiate between linear and circular convolution.

Part- II

Q2 Answer Any Eight out of Twelve

(06x08)

- Determine the energy and power of the signal $y[n] = 10 \cos\left(10n + \frac{\pi}{8}\right)$.
- Find the linear convolution of $x(n) = \{1, 4, 16, 32\}$ and $h(n) = \{1, 2, 3, 4\}$ without using the tabular method.
- Using the properties of Z transform, find the Z transform of the following
 - $x[n] = u[-n]$
 - $x[n] = 2^n u[n-2]$
- Determine the Z-transform of $x(n) = \{-3, -1, -8, 1, 2, 7, 3, 9, 15, 7\}$
- Find the cross correlation of $x(n) = \{1, -1, 1, -1\}$ and $h(n) = \{2, 2, 3, 3\}$
- Find the inverse Z-Transform of $H(z) = \frac{2z}{9z^2 - 16z + 1}$, $|z| > 1$ using long division method.
- Show that energy of power signal is ∞ .

- h) A causal system has input $\delta[n]$ and output $y[n]$. Find the frequency response and impulse response of the causal system if, $y[n] = \delta[n] + \frac{1}{6}\delta[n-1] - \frac{1}{6}\delta[n-2]$
- i) Write any four properties of DFT.
- j) How to find inverse z-transform using power series expansion method?
- k) Find the circular convolution of $x(n) = \{1, 2, 3\}$ & $h(n) = \{1, 1, 1\}$ using DFT method.
- l) Compute the DFT of $x[n] = \cos \frac{n\pi}{2}$ where $N = 4$.

Part-III

Answer Any Two out of Four

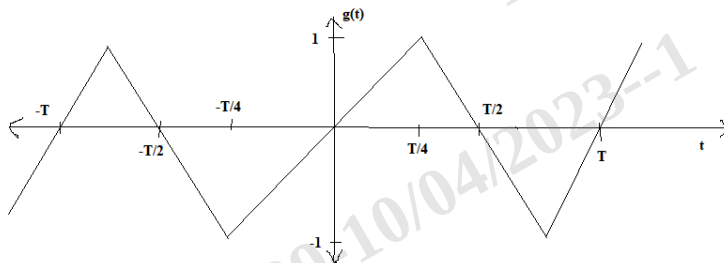
Q3 Determine the inverse z-transform of $X(z) = \frac{z}{-4z + 3z^2 + 1}$, if the ROC (16)

(A) $|z| > 1$

(B) $|z| < \frac{1}{3}$

(C) $1 > |z| > \frac{1}{3}$

Q4 Find the trigonometric Fourier series of the triangular waveform shown below in the following figure. (16)



Q5 Write notes of (any two) (16)

1. Properties of Continuous Time Fourier Transform
2. Discrete Time Systems Described by Difference Equations
3. Inverse Z-Transform Methods

Q6 Check the linearity, causality and time-invariance of the systems represented by (16)

(i) $g[n] = 12y[n] + \frac{30y[n]}{y[n-4]}$

(ii) $g[n] = F\{y[n]\} = 11y^2[n-5] + ny[n-1]$

Registration No.:

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Total Number of Pages: 02

Course: B.Tech, IDD (B.Tech and M.Tech)
Sub_Code: REC3C002

3rd Semester Regular/Back Examination: 2023-24

SUBJECT: Signals and Systems

BRANCH(S): ECE, ETC

Time: 3 Hour

Max Marks: 100

Q.Code: N508

Answer Question No.1 (Part-1) which is compulsory, any eight from Part-II and any two from Part-III.

The figures in the right hand margin indicate marks.

Part-I

Q1 Answer the following questions: (2 x 10)

- How to check stability of a discrete time system?
- Determine the DFT of $\delta(n) + \delta(n-1)$.
- $y(n) = x(n) + x(n-1)$ is time-invariant system or not.
- Differentiate between convolution and correlation of two discrete time signals.
- Write any two properties of discrete time convolution.
- Differentiate between linear and circular convolution.
- Write any two properties of z-transform.
- Write the formula for a continuous time Fourier Series expansion.
- What is circular convolution and how it is different from linear convolution?
- Define the superposition theorem to check linearity of a system.

Part-II

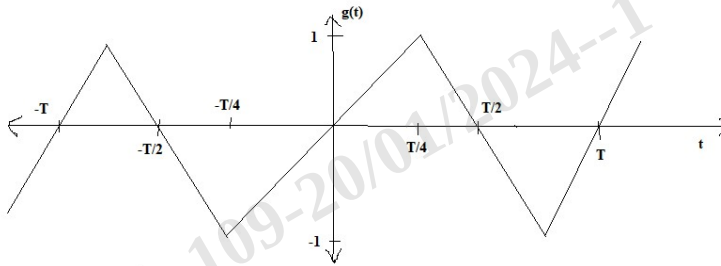
Q2 Only Focused-Short Answer Type Questions- (Answer Any Eight out of Twelve) (6 x 8)

- Determine the Z-transform of $x(n) = \{-3, 1, 2, 0, 0, 0, 1, 5, 9\}$
- Find the circular convolution of $x(n) = \{1, -1, 3\}$ & $h(n) = \{1, 0, -1\}$
- Check the linearity and time-invariance of the systems represented by

$$h[n] = y[n-1] + \frac{2y[n]}{y[n-3]}$$

- Determine the DFT of $2\delta(n) + 3\delta(n-1) + 3\delta(n-4)$.
- Find the z-transform of $x(n) = \cos \omega_0 n$ for $n \geq 0$.
- Using the properties of Z transform, find the Z transform of $3^n u[n-3]$.

- g) Find the fundamental term of the Fourier series of the waveform shown below.



- h) Find the inverse Z-transform of $H(z) = \frac{1 + 2z^{-1}}{1 - 4z^{-1} - 2z^{-2}}$.
- i) Find the DTFT of the sequence $x(n) = u(n - k)$.
- j) Derive the relationship between Fourier transform of $x(2t + 3)$ and that of $x(t)$.
- k) Find the linear convolution of $x(n) = \{2, 4, 6, 8\}$ and $h(n) = \{1, 2, 3, 4\}$ using the properties of Z-Transform.
- l) Why do we need both Fourier Series and Fourier transform to analyze the signals in frequency domain? Justify your answer with suitable examples.

Part-III

Only Long Answer Type Questions (Answer Any Two out of Four)

- Q3** Compute the DFT of $x[n] = \cos \frac{n\pi}{4} + \sin \frac{n\pi}{4}$ where $0 \leq n < 8$. (16)
- Q4** Find the inverse Z-Transform of $H(z) = \frac{z}{3z^2 - 4z + 1}$ using partial fraction expansion method for all possible ROCs. (16)
- Q5** Compute the continuous time Fourier transform of the signal (16)
- $$x(t) = \begin{cases} 1, & \text{for } -4 \leq t \leq 4 \\ 0, & \text{for } 4 \leq |t| \leq 10 \end{cases}$$
- Q6** Write Short Notes (Any Two) (8x2)
- Frequency Domain Sampling
Inverse Z-Transform
Properties of DFT